

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: CSA0409 COURSE CODE: Operating Systems

COURSE OUTCOME COVERED

***CO1:** Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments.*

QUESTIONS: 1

Total Marks:100

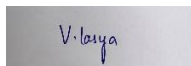
Annexure A

Technical Presentation

Code of Conduct:

I __V.LASYA(192524186)____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Technical Presentation [Any one]

Question 1

Suppose an Operating System is used in a banking system handling thousands of transactions simultaneously.

Prepare a technical presentation explaining:

Types of OS suitable for this scenario

OS structure used

Role of system calls in transaction processing

(BL3)



Operating System in a Banking System

Handling thousands of simultaneous transactions



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Presented By:
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The Scenario and What It Demands

1000s

transactions at the
same time

100%

accuracy: no lost or
duplicate money

24x7

availability and fast
response

Secure

authentication and
protected data

Why this matters:

Many customers deposit, withdraw and transfer at once, so the OS must schedule work fairly, protect shared account data and recover safely from failures.

Types of OS Suitable for This Scenario

OS type	Fit	Reason
Batch	Low	No interaction; unsuitable for live requests
Multiprogramming / Multitasking	High	Keeps CPU busy by running many requests together
Multi-user / Time-sharing	High	Many users share the system with quick response
Multiprocessor (parallel)	High	Several CPUs process transactions in parallel
Distributed (branches, ATMs)	High	Links branches and servers; gives reliability
Real-time (soft)	Partial	Needed for fast response, not hard deadlines

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Recommended OS: Multi-user Multitasking

Characteristics

- Multiprogramming and multitasking
- Multi-user with access control
- Multiprocessor support
- Virtual memory and journaled file system
- High availability and fault tolerance

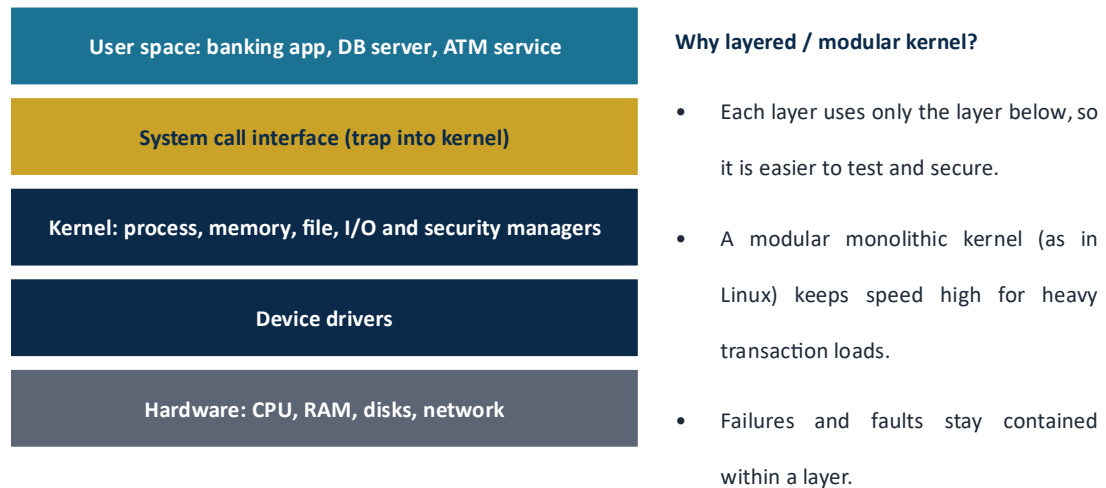
Examples in practice

- Linux and Unix servers
- IBM z/OS mainframes
- Windows Server

These run the database and transaction software that serves ATMs, net banking and branches.

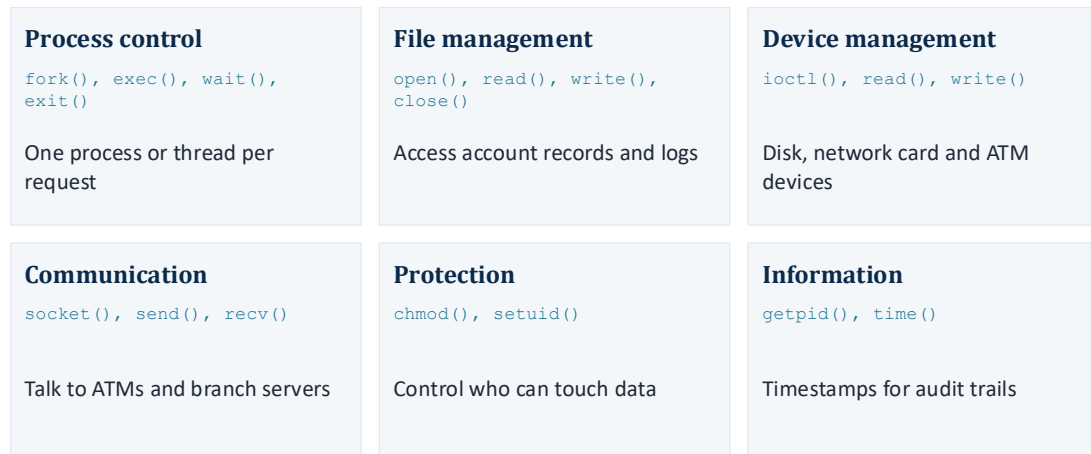
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OS Structure Used: Layered Kernel Design



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Role of System Calls in Transaction Processing



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A Transaction Through the OS



Example: transfer Rs. 5,000 from A to B. The OS locks both accounts, writes debit and credit, forces the log to disk, then unlocks. If power fails midway, the log lets the system roll back, so no money is lost.

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OS Support for Concurrency and Integrity

CPU scheduling

Round robin and priority scheduling share CPUs fairly across thousands of requests.

Synchronization

Mutexes and semaphores stop two transactions updating one account at once.

Memory management

Virtual memory and paging isolate processes and cache hot data.

Deadlock and recovery

Lock ordering and timeouts avoid deadlock; journaling restores state after a crash.

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Conclusion

- A multi-user, multitasking, multiprocessor server OS best fits banking.
- A layered, modular kernel gives security, speed and fault isolation.
- System calls are the controlled gateway to files, devices, networks and processes in every transaction.
- Together they deliver fast, safe and consistent transactions.

THANK YOU

RUBRICS FOR CALCULATION

Technical Presentation					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness